



**NATIONAL SKILL DEVELOPMENT  
CORPORATION**

**In Collaboration with  
YUVA INTERN**

**Virtual R Data Analyst Intern**

**WEEK 4 REPORT**

**Statistical Analysis and Predictive Modeling  
using R**

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# 1. INTRODUCTION

The four-week internship project focused on doing a data analysis of a Supermarket Sales Dataset using R. The project was set up as a continuous analytical process, taking the cleaned dataset from the first week and moving it through visualization, statistical analysis, and predictive modeling.

Week 1 focused on cleaning the data and initial exploration. The dataset was checked for missing values, entries, wrong data types, possible outliers, and variables that needed changing. Normalization, categorical encoding, and feature creation were also done.

Week 2 was about making visuals and sharing findings. Several visuals were made using R and ggplot2 to look at how sales were spread out, how different product lines did, how branches performed, sales over time, monthly patterns, customer satisfaction, and how customers bought things.

Week 3 took the analysis to a formal level with statistics and building models. Things like statistics looking at how data was spread, checking if data was normal, seeing how variables were connected, testing ideas using logistic regression, checking how good the models were using ten-fold cross-validation, looking at model problems, and making models better were all done.

Week 4 puts all those steps together into one report. The goal is not to go over the work again but to show how the different parts fit together and how the overall results can help with making decisions based on data and supporting business choices.

## 1.1 Overall Project Objective

The overall objective was to prepare a supermarket transaction dataset, identify patterns through visualization and statistical techniques, develop a predictive model for customer type, evaluate its performance, and communicate the results clearly to both technical and non-technical audiences.

## 1.2 Scope of the Final Report

In my view, the final report covers data preparation, exploratory analysis, visualization, statistical analysis, hypothesis testing, predictive modeling, model evaluation, model diagnostics, model comparison, business interpretation, challenges, lessons learned, and future scope.

# 2. DATASET OVERVIEW

The same Supermarket Sales Dataset was used throughout all four weeks. It contains 1,000 transaction records and 17 variables describing supermarket transactions, customers, products, sales, payment methods, ratings, and branch information.

| Attribute             | Value                     |
|-----------------------|---------------------------|
| Dataset Name          | Supermarket Sales Dataset |
| Rows                  | 1,000                     |
| Columns               | 17                        |
| Numerical Variables   | 9                         |
| Categorical Variables | 8                         |

The dataset contains transaction-level information such as branch, city, customer type, gender, product line, unit price, quantity, tax, sales, date, time, payment method, cost of goods sold, gross margin percentage, gross income, and customer rating.

## 2.1 Main Variables Used

| Variable           | Type        | Role / Description                                           |
|--------------------|-------------|--------------------------------------------------------------|
| Customer Type      | Categorical | Member or Normal; target variable for Week 3 classification. |
| Gender             | Categorical | Customer gender category.                                    |
| Product Line       | Categorical | Product category purchased.                                  |
| Unit Price         | Numerical   | Price of one unit.                                           |
| Quantity           | Numerical   | Number of units purchased.                                   |
| Sales              | Numerical   | Total transaction sales value.                               |
| Rating             | Numerical   | Customer rating.                                             |
| Payment            | Categorical | Payment method.                                              |
| Branch / City      | Categorical | Location information.                                        |
| Month / Day / Hour | Derived     | Time features created for analysis and modeling.             |

```
[4]: dim(sales)

1000 · 17
```

Figure 1: Dataset dimension output showing 1,000 observations and 17 variables.

## 3. METHODOLOGY

The project followed a sequential analytical workflow. Each stage prepared the output required by the next stage, allowing the four weeks to form one coherent project.

DATA INSPECTION → DATA CLEANING → TRANSFORMATION & FEATURE ENGINEERING → VISUALIZATION → STATISTICAL ANALYSIS → HYPOTHESIS TESTING → LOGISTIC REGRESSION → MODEL EVALUATION → CROSS-VALIDATION → DIAGNOSTICS → MODEL COMPARISON → INTERPRETATION

### 3.1 Week-wise Integration

| Week   | Primary Task                                 | Contribution to Final Project                                              |
|--------|----------------------------------------------|----------------------------------------------------------------------------|
| Week 1 | Data Cleaning and Preliminary Analysis       | Created a consistent and analysis-ready dataset.                           |
| Week 2 | Data Visualization and Insight Communication | Converted transaction data into interpretable visual patterns.             |
| Week 3 | Statistical Analysis and Predictive Modeling | Tested relationships and developed/evaluated a classification model.       |
| Week 4 | Comprehensive Reporting and Presentation     | Integrated all results, discussed implications, and presented conclusions. |

## 3.2 Tools and Packages

R was used as the primary programming language. The Week 2 visualization work used ggplot2, dplyr, and scales. Statistical analysis and modeling used standard R statistical functions and the logistic regression glm() function.

## 4. DATA PREPARATION AND CLEANING

The first stage ensured that the dataset was reliable and suitable for later analysis. The Week 1 report confirmed that the dataset contained no missing values and no duplicate records. Data types were checked, column names were standardized, potential outliers were reviewed, and additional variables were created.

### 4.1 Missing Values

Missing values were checked using is.na(). No missing values were found in the dataset, so no imputation method was required.

### 4.2 Duplicate Records

Duplicate records were checked using duplicated(). No duplicate records were detected, indicating that the dataset did not require duplicate-row removal.

### 4.3 Data Type Verification and Column Names

The structure of the dataset was checked using str(). Numerical variables were verified as numeric, and categorical variables were treated appropriately. Column names were standardized by converting them to lowercase and replacing spaces with underscores, improving consistency and readability in R.

```
[6]: str(sales)
'data.frame':  1000 obs. of  17 variables:
 $ Invoice.ID      : chr  "750-67-8428" "226-31-3081" "631-41-3108" "123-19-1176" ...
 $ Branch         : chr  "Alex" "Giza" "Alex" "Alex" ...
 $ City           : chr  "Yangon" "Naypyitaw" "Yangon" "Yangon" ...
 $ Customer.type  : chr  "Member" "Normal" "Normal" "Member" ...
 $ Gender         : chr  "Female" "Female" "Female" "Female" ...
 $ Product.line   : chr  "Health and beauty" "Electronic accessories" "Home and lifestyle" "Health and beauty" ...
 $ Unit.price     : num  74.7 15.3 46.3 58.2 86.3 ...
 $ Quantity       : int   7 5 7 8 7 7 6 10 2 3 ...
 $ Tax.S.        : num  26.14 3.82 16.22 23.29 30.21 ...
 $ Sales          : num  549 80.2 340.5 489 634.4 ...
 $ Date          : chr   "1/5/2019" "3/8/2019" "3/3/2019" "1/27/2019" ...
 $ Time          : chr   "1:08:00 PM" "10:29:00 AM" "1:23:00 PM" "8:33:00 PM" ...
 $ Payment       : chr   "Ewallet" "Cash" "Credit card" "Ewallet" ...
 $ cogs          : num  522.8 76.4 324.3 465.8 604.2 ...
 $ gross.margin.percentage: num  4.76 4.76 4.76 4.76 4.76 ...
 $ gross.income  : num  26.14 3.82 16.22 23.29 30.21 ...
 $ Rating        : num  9.1 9.6 7.4 8.4 5.3 4.1 5.8 8 7.2 5.9 ...
```

Figure 2: R structure output showing the 1,000 observations and 17 variables with their data types.

### 4.4 Outlier Detection

Outliers were examined using boxplots and the Interquartile Range (IQR) approach. Some high-value observations were identified as possible outliers. However, these observations represented legitimate high-value transactions rather than data-entry errors, so they were retained.

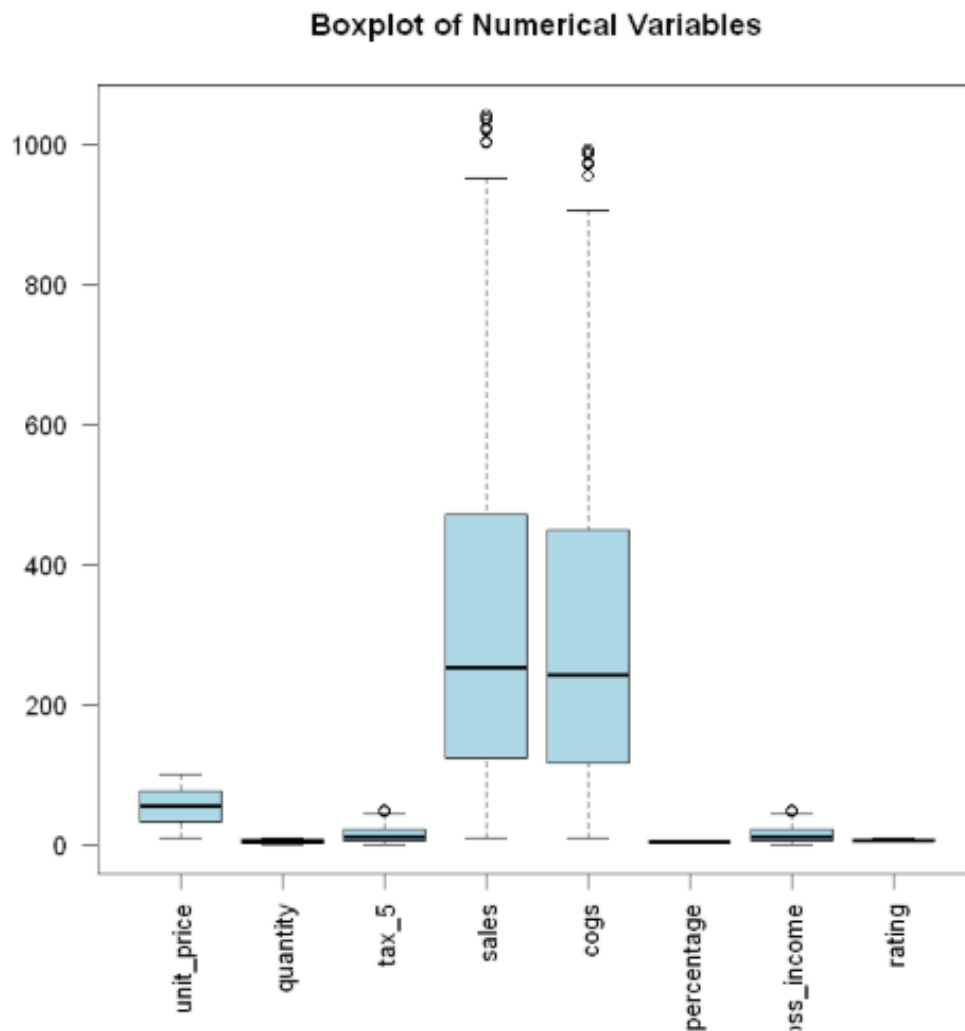


Figure 3: Boxplot-based outlier detection for numerical variables.

## 4.5 Data Transformation

Sales values were standardized using Z-score normalization. Categorical variables such as gender, customer type, and payment method were encoded into numerical representations where required for analysis and modeling.

### 8. Encoding Categorical Variables

Categorical variables are encoded into numeric representations. This preprocessing step is useful for machine learning models that require numerical input.

```
[19]: sales$gender_encoded <- as.numeric(as.factor(sales$gender))
      sales$payment_encoded <- as.numeric(as.factor(sales$payment))
      sales$customer_type_encoded <- as.numeric(as.factor(sales$customer_type))

      head(sales[, c("gender", "gender_encoded",
                    "payment", "payment_encoded",
                    "customer_type", "customer_type_encoded")])
```

A data.frame: 6 × 6

|   | gender | gender_encoded | payment     | payment_encoded | customer_type | customer_type_encoded |
|---|--------|----------------|-------------|-----------------|---------------|-----------------------|
|   | <chr>  | <dbl>          | <chr>       | <dbl>           | <chr>         | <dbl>                 |
| 1 | Female | 1              | Ewallet     | 3               | Member        | 1                     |
| 2 | Female | 1              | Cash        | 1               | Normal        | 2                     |
| 3 | Female | 1              | Credit card | 2               | Normal        | 2                     |
| 4 | Female | 1              | Ewallet     | 3               | Member        | 1                     |
| 5 | Female | 1              | Ewallet     | 3               | Member        | 1                     |
| 6 | Female | 1              | Ewallet     | 3               | Member        | 1                     |

Figure 4: Encoding of categorical variables.

## 4.6 Feature Engineering

The Date and Time fields were used to create analytic variables. These extra analytic variables include month, day, hour, and time period information. These derived variables were later used in Week 2 time-based visualizations and Week 3 modeling.

## 4.7 Preliminary Correlation Analysis

Pearson correlation analysis was performed on the variables. I noticed positive relationships among sales, COGS, gross income, and tax. This is expected because sales, COGS, gross income, and tax are mathematically related to transaction value and cost.

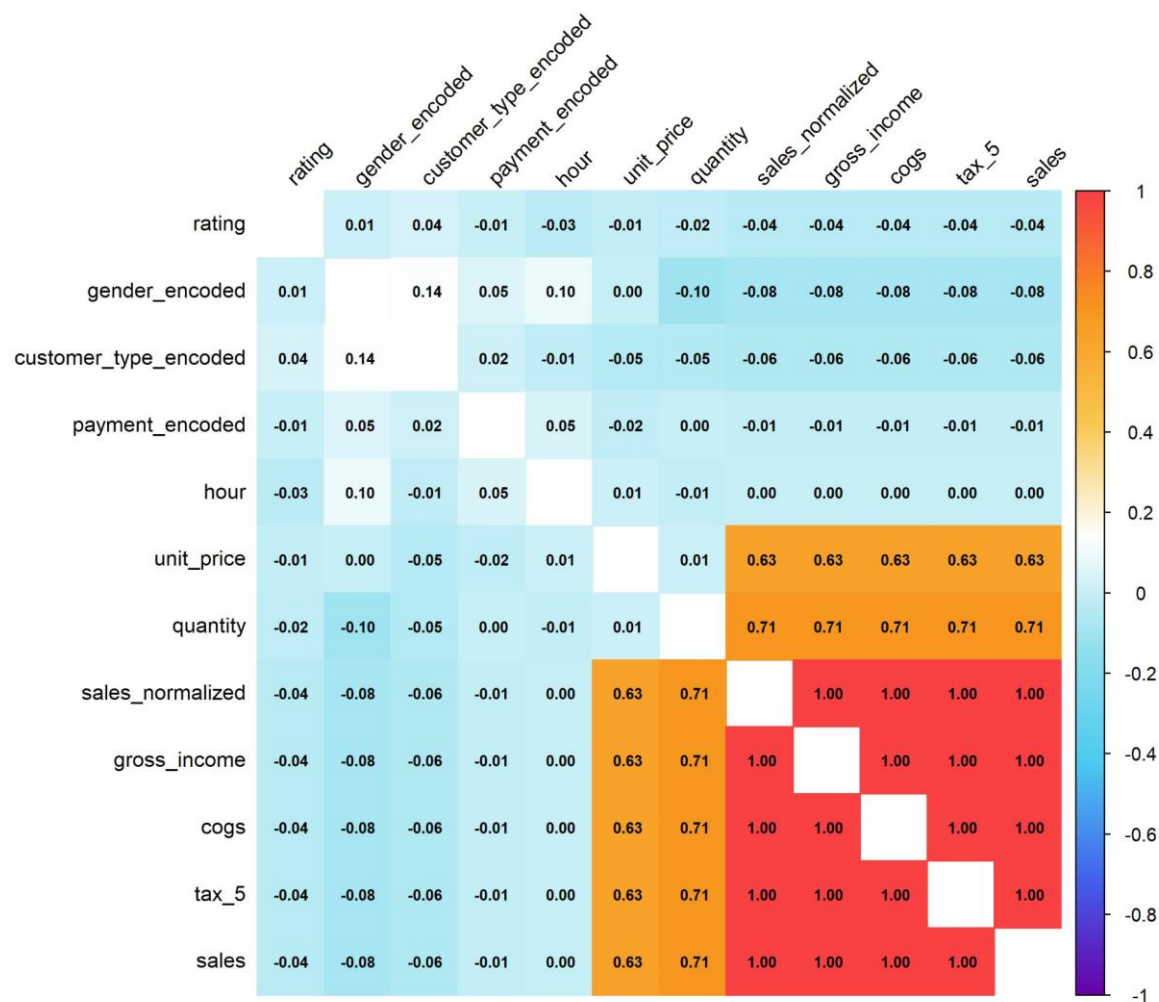


Figure 5: Week 1 correlation heatmap.

## 4.8 Data Preparation Outcome

After I finished the preparation stage, I had a dataset. This dataset had no missing values and no duplicate records. I kept the names and types in a standard form. I made sure to keep the real high-value transactions. I normalized the sales amounts. I encoded the variables. I added time-based features. Then I carried this prepared dataset into Weeks 2 and 3.

## 5. DATA VISUALIZATION AND EXPLORATORY ANALYSIS

In Week 2, I worked on turning the prepared transaction data into pictures that anyone could understand. I chose kinds of charts based on the specific questions I wanted to answer. The findings below bring together the seven visualizations created during Week 2.

### 5.1 Distribution of Supermarket Sales

I looked at a histogram to see how our transaction sales are spread out. Most of the transaction sales fall into the 0 to 1,000 range. The mean sales value came out to 322.97. The median was a bit lower at 253.85. Because the mean is higher than the median, it shows that those big transaction sales are pulling the average up.

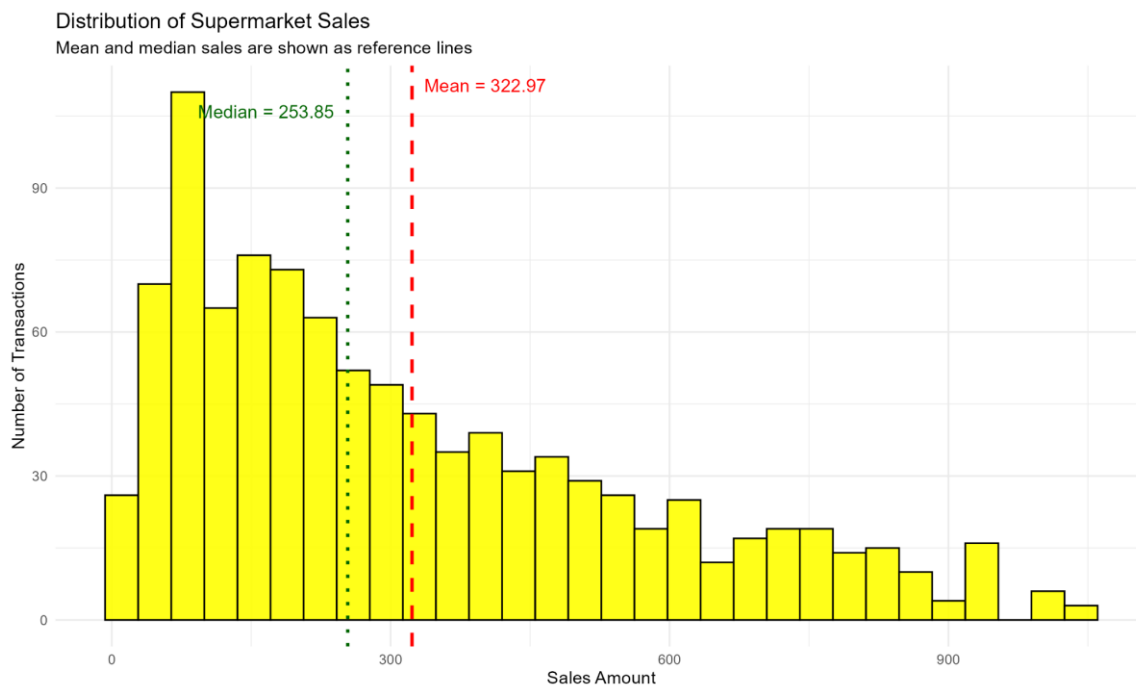


Figure 6: Distribution of Supermarket Sales.

Business insight: The distribution provides an understanding of typical transaction values and highlights the presence of relatively high-value purchases that may require separate consideration.

### 5.2 Product Line Performance

A lollipop chart showed sales across different product lines. The product line that performed the best had sales of 56,145. The product line that performed the worst had sales of 49,194. This means that each product category did not add the same amount to total sales. The product categories did not contribute equally to sales.



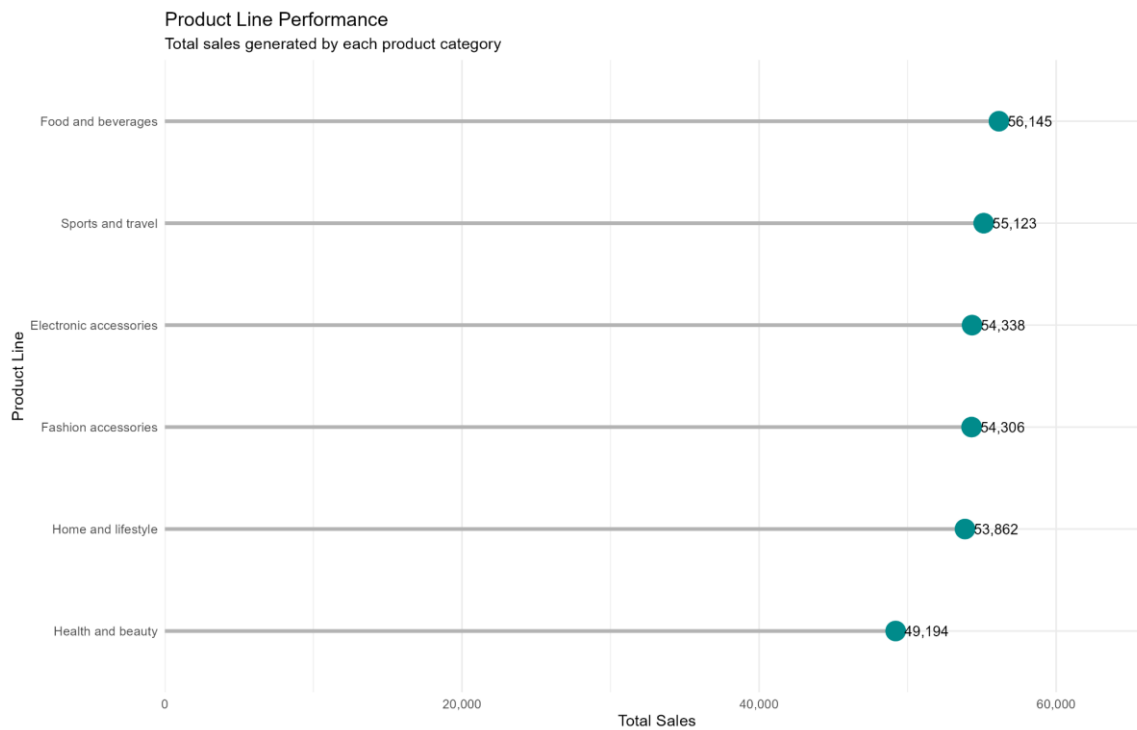


Figure 7: Product Line Performance.

Business insight: Higher-performing product categories may require continued inventory support and effective promotion, while lower-performing categories can be investigated for demand and sales opportunities.

### 5.3 Branch Performance Overview

I looked at a bubble chart that compares branches using total sales, average customer rating, and transaction volume. In the Week 2 analysis, Giza had the highest total sales, and Giza also had the highest average customer rating. This chart shows why we should not look at sales when we evaluate how a branch is performing.

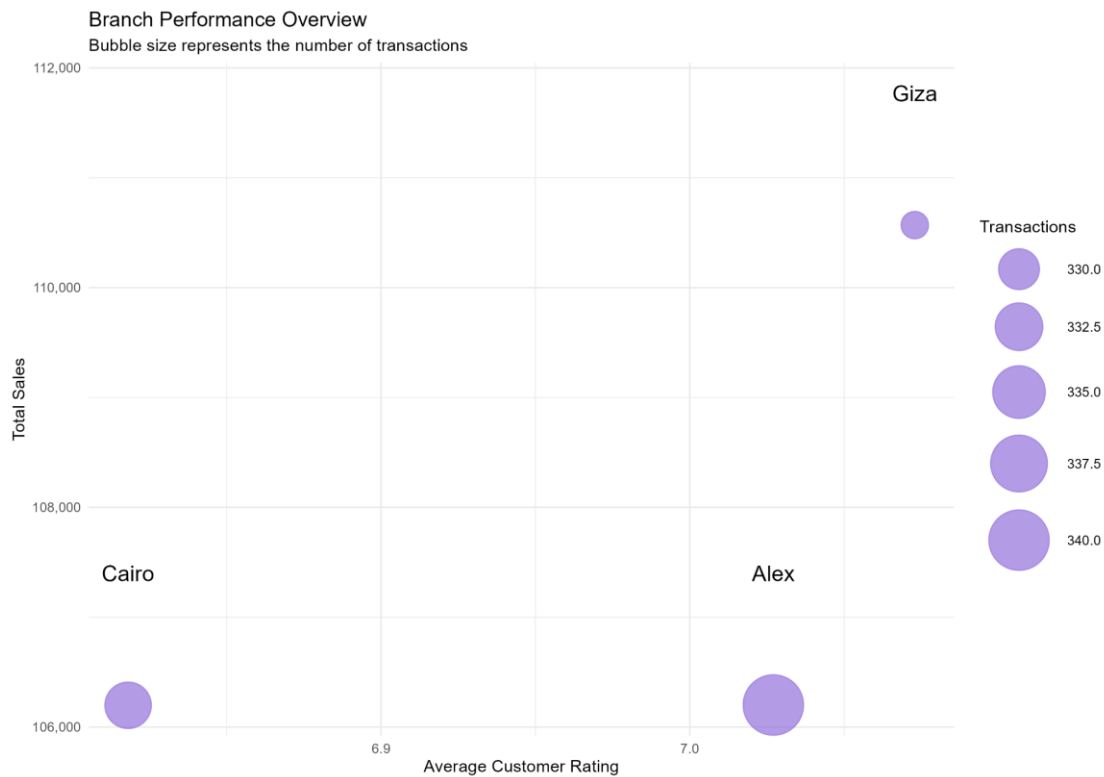


Figure 8: Branch Performance Overview.

Business insight: Combining revenue, customer satisfaction, and transaction volume gives management a broader view of branch performance.

## 5.4 Sales Intensity by Day and Hour

A heatmap was used to examine sales intensity across days and hours. The highest sales intensity occurred on Tuesday at 19:00 (7 PM), while relatively low sales activity occurred on Monday at 14:00 (2 PM).

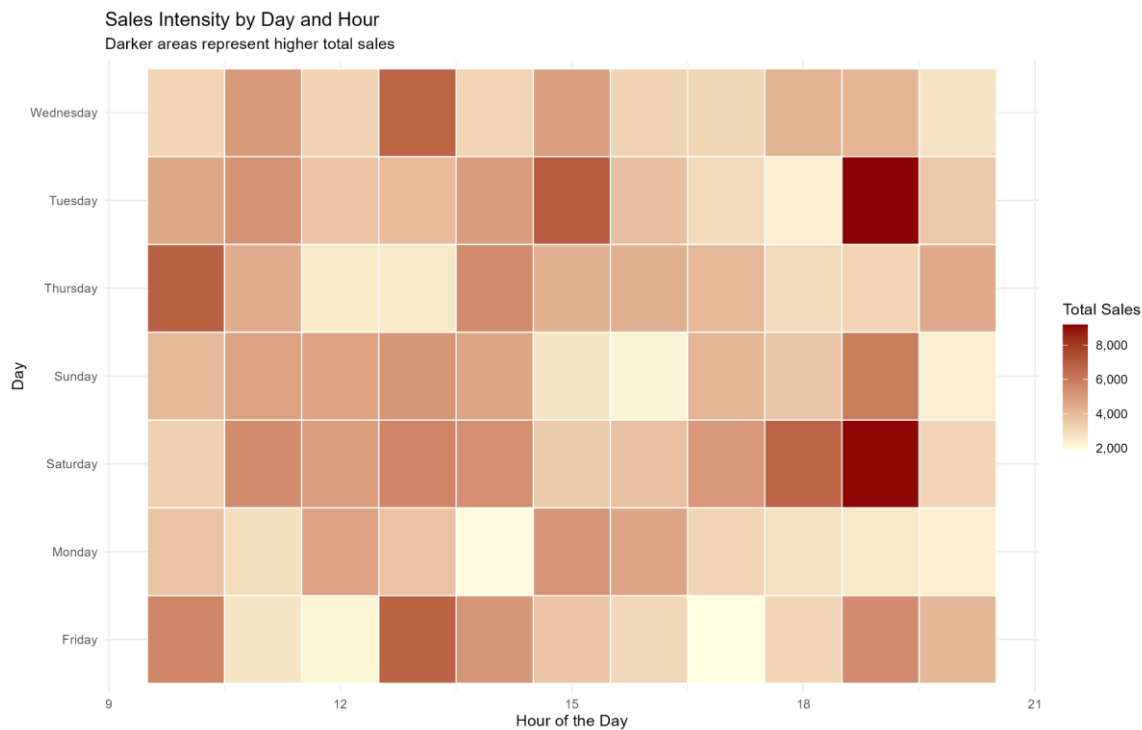


Figure 9: Sales Intensity by Day and Hour.

Business insight: Identifying peak periods can support staff scheduling, inventory preparation, and customer-service planning.

## 5.5 Monthly Sales Trend

The monthly line chart showed fluctuations in total sales across the months in the dataset. January recorded the highest total sales at 116,292, while February recorded the lowest at 97,216. Sales therefore did not remain constant over time.

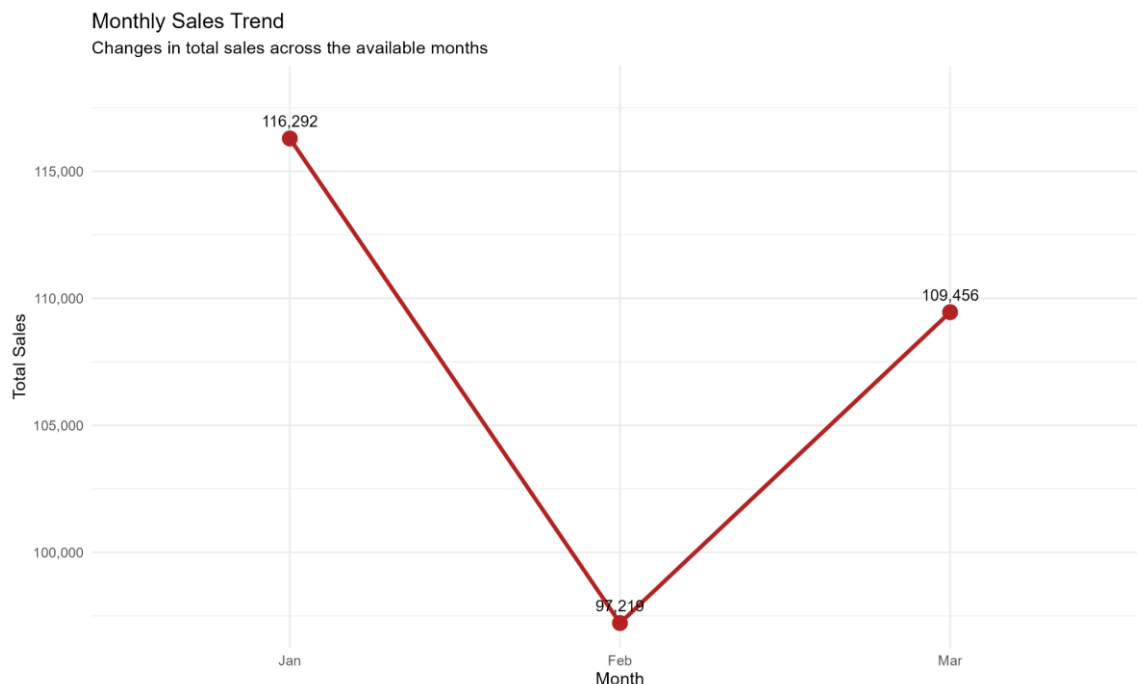


Figure 10: Monthly Sales Trend.

Business insight: Monthly variation can be considered when planning inventory, promotions, staffing, and sales targets.

## 5.6 Customer Satisfaction Across Product Lines

Average customer ratings varied by product line. Food and beverages recorded the highest average rating at 7.11, while Home and lifestyle recorded the lowest at 6.84. The results show that the product line with stronger sales is not necessarily the product line with the highest customer satisfaction.

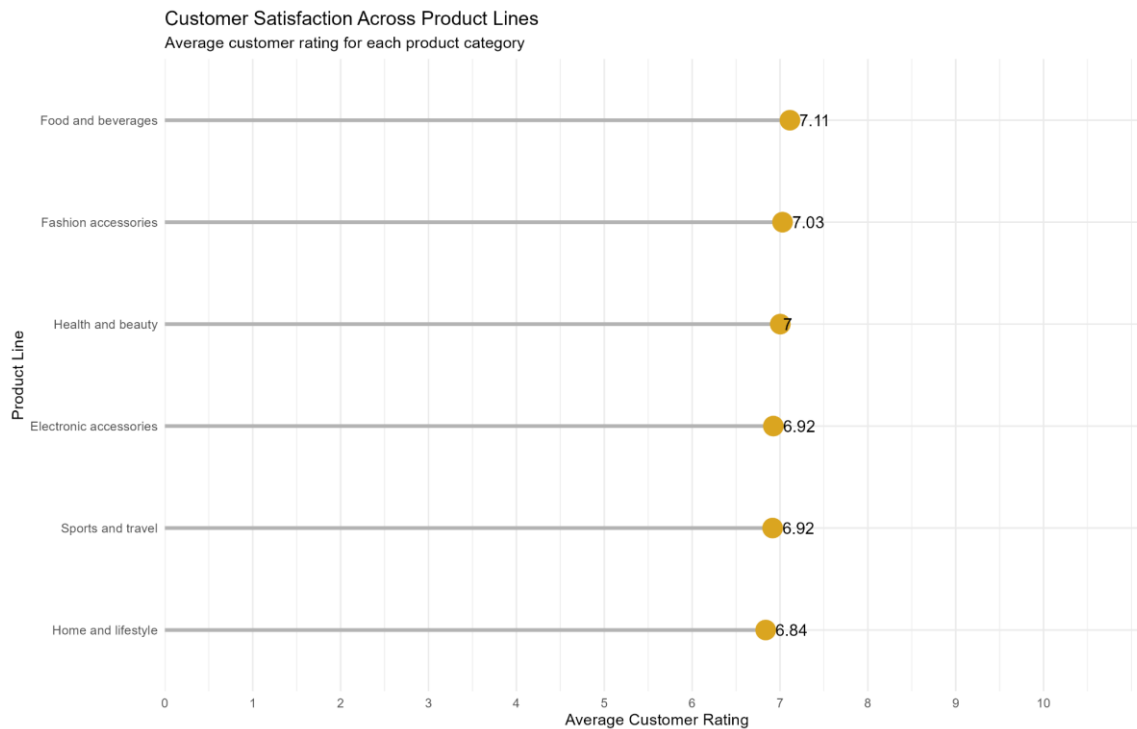


Figure 11: Customer Satisfaction Across Product Lines.

Business insight: Product performance should be evaluated using both financial outcomes and customer feedback.

## 5.7 Sales Distribution by Customer Type

The violin plot compared sales distributions for Member and Normal customers. Member customers had a higher mean sales value and showed slightly greater variation in transaction values.



Figure 12: Sales Distribution by Customer Type.

Business insight: Differences between Member and Normal customers can support customer-retention strategies, targeted promotions, and personalized marketing.

## 5.8 Overall Visualization Findings

- Transactions mainly fell in the lower to middle price ranges. Some transactions were higher value.
- Product lines each added an amount to total sales.
- Branch performance changed when sales, customer ratings, and transaction volume were all considered together.
- Sales activity changed by day and hour with peaks and valleys.
- Monthly sales fluctuated; January had the highest total sales, while February had the lowest.
- Customer satisfaction varied from one product line to another.
- Member customers and Normal customers had sales distributions; Member customers had a higher average sales value.

## 6. STATISTICAL ANALYSIS

Week 3, we solidified the patterns we found in our analysis. We calculated statistics for unit price, quantity, rating, month, day, and hour. By using histograms to examine distributions, we then carried out normality testing, correlation analysis, and hypothesis testing.

### 6.1 Descriptive Statistics

From the analysis, unit price ranged from about 10.08 to 199.96. Quantity varied between 1 and 10, and customer ratings spanned from 4.0 to 10.0. The average quantity was 5.51, and the average rating was about 6.97.

| unit_price    | quantity      | rating         | month      |
|---------------|---------------|----------------|------------|
| Min. :10.08   | Min. : 1.00   | Min. : 4.000   | Jan :352   |
| 1st Qu.:32.88 | 1st Qu.: 3.00 | 1st Qu.: 5.500 | Mar :345   |
| Median :55.23 | Median : 5.00 | Median : 7.000 | Feb :303   |
| Mean :55.67   | Mean : 5.51   | Mean : 6.973   | Apr : 0    |
| 3rd Qu.:77.94 | 3rd Qu.: 8.00 | 3rd Qu.: 8.500 | May : 0    |
| Max. :99.96   | Max. :10.00   | Max. :10.000   | Jun : 0    |
|               |               |                | (Other): 0 |

| day          | hour          |
|--------------|---------------|
| Length :1000 | Min. :10.00   |
| N.unique : 7 | 1st Qu.:12.00 |
| N.blank : 0  | Median :15.00 |
| Min.nchar: 6 | Mean :14.91   |
| Max.nchar: 9 | 3rd Qu.:18.00 |
|              | Max. :20.00   |

Figure 13: Week 3 descriptive statistics output.

## 6.2 Distribution Analysis

Histograms were created for quantity, unit price, and customer rating to examine their distributions before applying statistical tests.

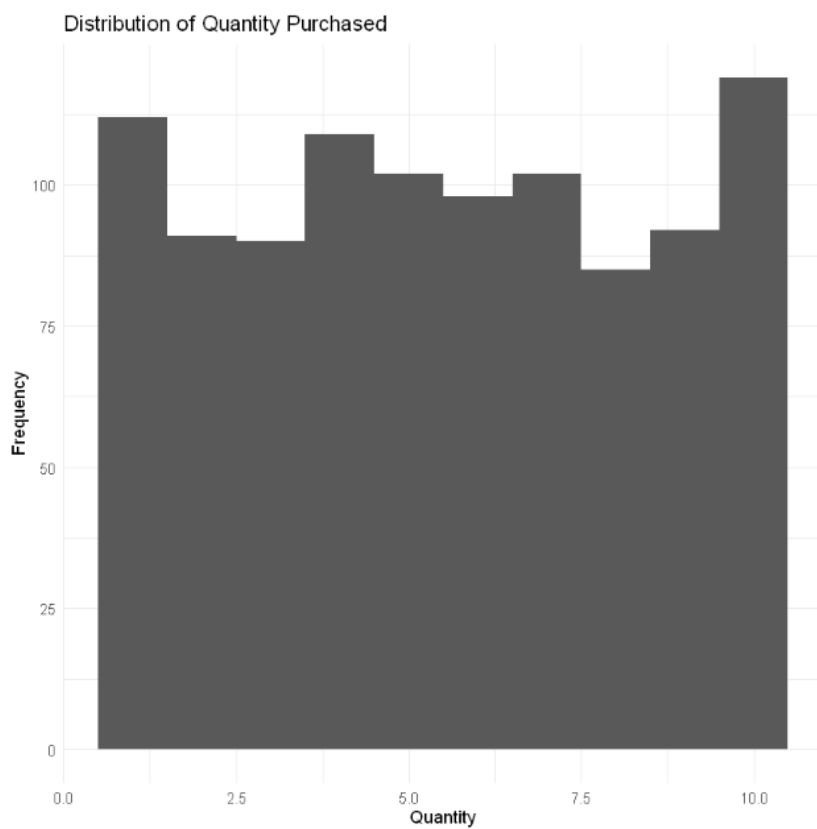


Figure 14: Distribution of Quantity Purchased.

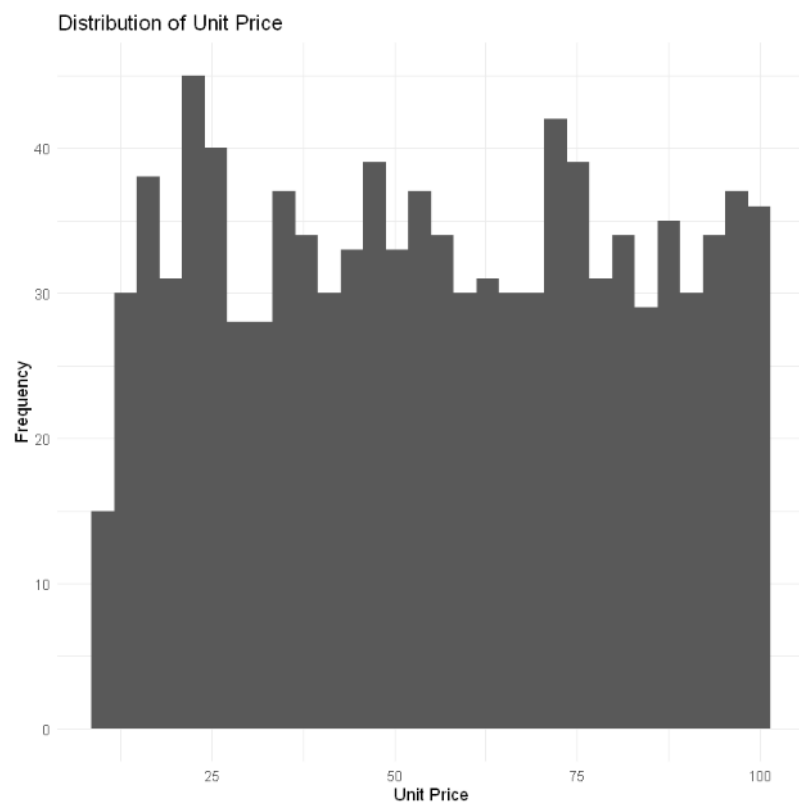


Figure 15: Distribution of Unit Price.

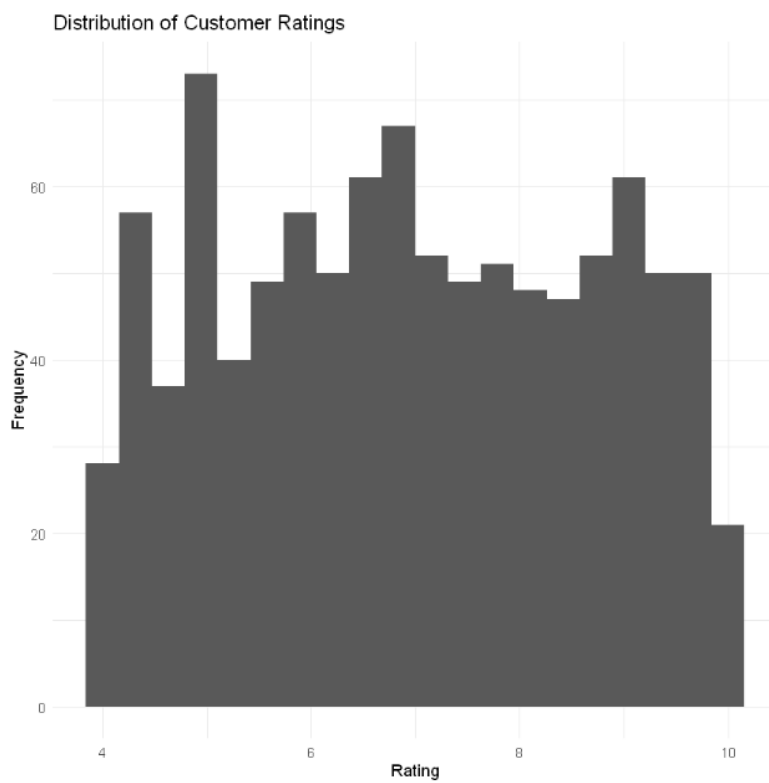


Figure 16: Distribution of Customer Ratings.

### 6.3 Normality Testing

The Shapiro-Wilk test was used to check if quantity, unit price, and rating had a normal distribution. For each of the three variables, the p-values were less than 0.05. This gave evidence that the normal distribution was not a fit. The null hypothesis of normality was rejected.

#### Shapiro-Wilk normality test

```
data: model_data$quantity  
W = 0.93258, p-value < 2.2e-16
```

#### Shapiro-Wilk normality test

```
data: model_data$unit_price  
W = 0.95188, p-value < 2.2e-16
```

#### Shapiro-Wilk normality test

```
data: model_data$rating  
W = 0.9582, p-value = 2.688e-16
```

Figure 17: Shapiro-Wilk normality test results.

| Variable   | W statistic | p-value   | Interpretation             |
|------------|-------------|-----------|----------------------------|
| Quantity   | 0.93258     | < 2.2e-16 | Evidence against normality |
| Unit Price | 0.95188     | < 2.2e-16 | Evidence against normality |
| Rating     | 0.95820     | 2.688e-16 | Evidence against normality |

### 6.4 Correlation Analysis

I ran a Pearson correlation analysis on all the numeric variables in the modeling dataset. The correlation matrix gave me a look at how unit price, quantity, rating, and time-related numeric variables all connect.



'unit\_price' · 'quantity' · 'rating' · 'hour'

A matrix: 4 × 4 of type dbl

|            | unit_price   | quantity     | rating       | hour         |
|------------|--------------|--------------|--------------|--------------|
| unit_price | 1.000000000  | 0.010777564  | -0.008777507 | 0.008242210  |
| quantity   | 0.010777564  | 1.000000000  | -0.015814905 | -0.007316886 |
| rating     | -0.008777507 | -0.015814905 | 1.000000000  | -0.030587644 |
| hour       | 0.008242210  | -0.007316886 | -0.030587644 | 1.000000000  |

Figure 18: Correlation matrix for selected numerical variables.

## 6.5 Hypothesis Testing

Three hypothesis tests were conducted to examine whether customer type was associated with selected transaction characteristics.

### 6.5.1 T-Test: Customer Type and Quantity

The two-sample Welch t-test tested whether the mean quantity purchased differed between Member and Normal customers.

H<sub>0</sub>: There is no significant difference in mean quantity purchased between Member and Normal customers.

H<sub>1</sub>: There is a significant difference in mean quantity purchased between Member and Normal customers.

The test produced  $t = 1.4867$ ,  $df = 945.97$ , and  $p = 0.1374$ . Since  $p > 0.05$ , the null hypothesis was not rejected. Therefore, the analysis did not find statistically significant evidence of a difference in mean quantity purchased between the two customer groups.

```
Welch Two Sample t-test

data: quantity by customer_type
t = 1.4867, df = 945.97, p-value = 0.1374
alternative hypothesis: true difference in means between group Member and group Normal is not equal to 0
95 percent confidence interval:
 -0.08833615  0.64046717
sample estimates:
mean in group Member mean in group Normal
      5.630088          5.354023
```

Figure 19: Welch Two Sample t-test output.

### 6.5.2 Chi-Square Test: Customer Type and Gender

The chi-square test of independence examined whether customer type and gender were associated.

H<sub>0</sub>: Customer type and gender are independent.

H<sub>1</sub>: Customer type and gender are associated.

The test produced  $\chi^2 = 17.962$  with 1 degree of freedom and  $p = 2.253e-05$ . Since  $p < 0.05$ , the null hypothesis was rejected. The results provide evidence of an association between customer type and gender in this dataset.

### Pearson's Chi-squared test with Yates' continuity correction

```
data: gender_table  
X-squared = 17.962, df = 1, p-value = 2.253e-05
```

Figure 20: Chi-square test for customer type and gender.

#### 6.5.3 Chi-Square Test: Customer Type and Payment

A second chi-square test examined whether customer type and payment method were associated.

H<sub>0</sub>: Customer type and payment method are independent.

H<sub>1</sub>: Customer type and payment method are associated.

The test produced  $\chi^2 = 2.6714$  with 2 degrees of freedom and  $p = 0.263$ . Since  $p > 0.05$ , the null hypothesis was not rejected. The analysis therefore did not find statistically significant evidence of an association between customer type and payment method.

### Pearson's Chi-squared test

```
data: payment_table  
X-squared = 2.6714, df = 2, p-value = 0.263
```

Figure 21: Chi-square test for customer type and payment method.

#### 6.6 Statistical Summary

| Analysis                             | Result          | Conclusion                           |
|--------------------------------------|-----------------|--------------------------------------|
| Normality: Quantity                  | $p < 2.2e-16$   | Evidence against normality.          |
| Normality: Unit Price                | $p < 2.2e-16$   | Evidence against normality.          |
| Normality: Rating                    | $p = 2.688e-16$ | Evidence against normality.          |
| T-test: Customer Type vs Quantity    | $p = 0.1374$    | No significant difference detected.  |
| Chi-square: Customer Type vs Gender  | $p = 2.253e-05$ | Significant association detected.    |
| Chi-square: Customer Type vs Payment | $p = 0.263$     | No significant association detected. |

## 7. PREDICTIVE MODELING

The predictive modeling stage focused on classifying customers as Member or Normal. Because customer\_type is a binary categorical variable, logistic regression was selected as the primary classification method.

## 7.1 Model Selection

We used gender, product line, unit price, quantity, rating, payment, month, day, and hour as the predictors for the logistic regression model. I decided to leave out the variable. This is because customer\_type\_encoded holds the info, as our target, and keeping customer\_type\_encoded would lead to information leakage.

## 7.2 Train-Test Split

The data was split into two parts, one for training and one for testing. The split was done in a way that 80 percent of the data was for training and 20 percent was for testing. A specific random seed number 123 was used so that the split could be repeated if needed. The part that was used for training was used to build the model. The part that was used for testing was used to check how well the model worked.

```
# Part 14 – Train/Test Split

set.seed(123)

# Create random training indices
train_index <- sample(
  1:nrow(model_data),
  size = 0.80 * nrow(model_data)
)

# Create training data
train_data <- model_data[train_index, ]

# Create testing data
test_data <- model_data[-train_index, ]

# Check sizes
nrow(train_data)
nrow(test_data)
```

800

200

Figure 22: Week 3 train-test split output.

## 7.3 Logistic Regression

The logistic regression model estimated the probability that a transaction belonged to one of the customer-type categories. Predicted probabilities were converted into predicted classes using a 0.5 classification threshold.

```

Call:
glm(formula = customer_type ~ gender + product_line + quantity +
    payment + rating, family = binomial, data = train_data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.341169   0.399728  -0.854 0.393381
genderMale      0.557039   0.146765   3.795 0.000147 ***
product_lineFashion accessories  0.001173   0.244044   0.005 0.996164
product_lineFood and beverages -0.223640   0.243319  -0.919 0.358031
product_lineHealth and beauty  -0.057194   0.257104  -0.222 0.823960
product_lineHome and lifestyle -0.109423   0.255059  -0.429 0.667914
product_lineSports and travel  -0.139653   0.251419  -0.555 0.578582
quantity       -0.031062   0.025306  -1.227 0.219658
paymentCredit card  -0.248527   0.180769  -1.375 0.169184
paymentEwallet      0.120662   0.173240   0.697 0.486114
rating           0.021845   0.042784   0.511 0.609633
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1098.4  on 799  degrees of freedom
Residual deviance: 1073.9  on 789  degrees of freedom
AIC: 1095.9

Number of Fisher Scoring iterations: 4

```

Figure 23: Logistic regression model summary output.

## 7.4 Confusion Matrix

The confusion matrix compared actual customer types with predicted customer types. It showed 90 Member observations correctly classified as Member, 29 Member observations classified as Normal, 57 Normal observations classified as Member, and 24 Normal observations correctly classified as Normal.

|        | Predicted |        |
|--------|-----------|--------|
| Actual | Member    | Normal |
| Member | 90        | 29     |
| Normal | 57        | 24     |

Figure 24: Confusion matrix for the full logistic regression model.

## 7.5 Model Performance

| Metric   | Result | Interpretation                                              |
|----------|--------|-------------------------------------------------------------|
| Accuracy | 57.00% | 57% of test observations were correctly classified overall. |

|             |        |                                                                                  |
|-------------|--------|----------------------------------------------------------------------------------|
| Sensitivity | 75.63% | The model correctly identified a relatively high proportion of Member customers. |
| Specificity | 29.63% | The model was less effective at correctly identifying Normal customers.          |

The model therefore performed better at recognizing Member customers compared to other customers. With an accuracy of 57 percent, the model gives helpful initial signs but is not accurate enough to be used as a complete system for classifying customers.

## 7.6 Ten-Fold Cross-Validation

I ran a tenfold cross-validation on the training data to see if the model stays consistent when looking at different parts of the data. The mean cross-validation accuracy ended up being 56.75%. I noticed that the individual fold accuracies changed quite a bit. This tells me that the model performance varies depending on which subset of the training data the model uses.

```
[79]: mean_cv_accuracy <- mean(cv_accuracy)
      mean_cv_accuracy
      0.5675

[80]: mean_cv_accuracy_percentage <- mean_cv_accuracy * 100
      mean_cv_accuracy_percentage
      56.75
```

Figure 25: Ten-fold cross-validation accuracy output.

```
[78]: # Part 23 - 10-Fold Cross-Validation
      set.seed(123)
      k <- 10
      folds <- sample(
        rep(1:k, length.out = nrow(train_data))
      )
      cv_accuracy <- numeric(k)
      for (i in 1:k) {
        cv_train <- train_data[folds != i, ]
        cv_test <- train_data[folds == i, ]
        cv_model <- glm(
          customer_type ~ gender +
            product_line +
            unit_price +
            quantity +
            rating +
            payment +
            month +
            day +
            hour,
          data = cv_train,
          family = binomial
        )
        cv_probability <- predict(
          cv_model,
          newdata = cv_test,
          type = "response"
        )
        cv_prediction <- ifelse(
          cv_probability >= 0.5,
          "Normal",
          "Member"
        )
        cv_accuracy[i] <- mean(
          cv_prediction == cv_test$customer_type
        )
      }
      cv_accuracy
      0.4875 0.5875 0.55 0.6375 0.6125 0.6375 0.55 0.5875 0.5625 0.4625
```

Figure 26: Individual cross-validation fold accuracies.

## 7.7 Model Diagnostics

Standard diagnostic plots were generated for the fitted logistic regression model. These plots were used to inspect residual behavior, leverage, and other aspects of model fit.

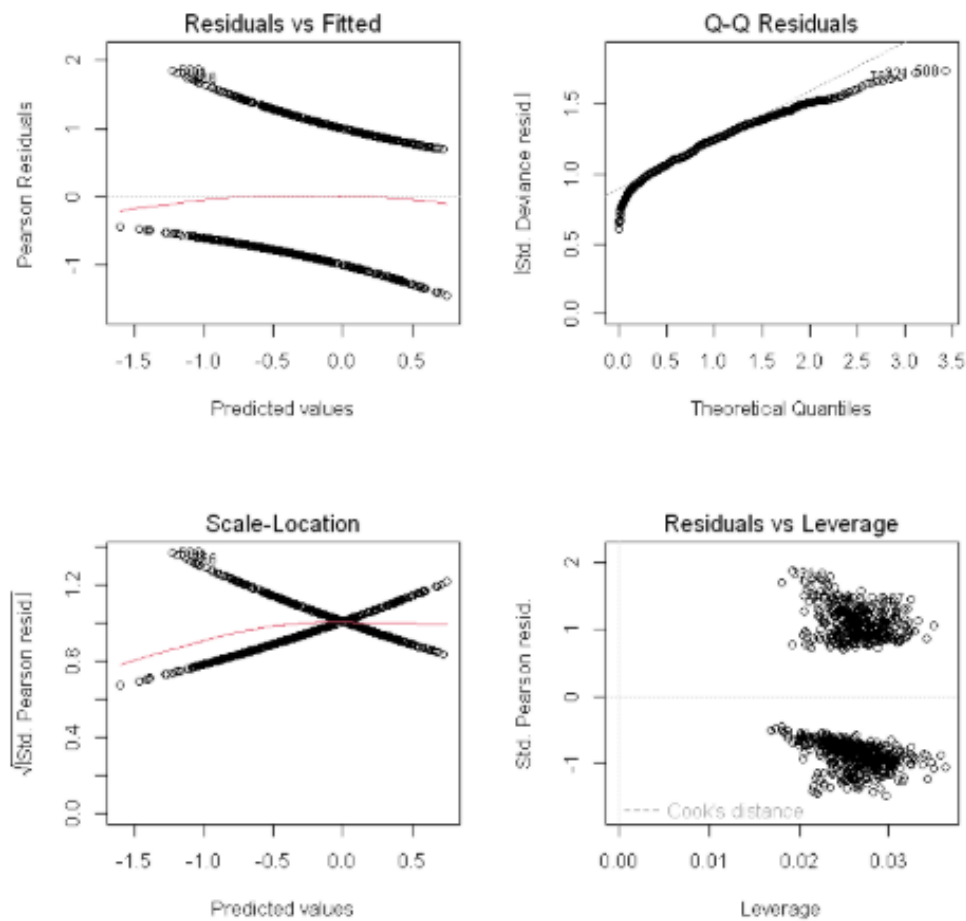


Figure 27: Logistic regression diagnostic plots.

## 7.8 Model Optimization

A second logistic regression model was created using gender, product line, quantity, payment, and rating. The goal was to see if a simpler set of predictors could keep or increase the accuracy of classification.

The better model got 58 percent test accuracy, while the full model got 57 percent. The reduced model increased test accuracy by one percentage point and used fewer predictors.

```
improved_accuracy <- sum(
  diag(improved_confusion)
) / sum(improved_confusion)

improved_accuracy
```

0.58

```
improved_accuracy_percentage <- improved_accuracy * 100

improved_accuracy_percentage
```

58

Figure 28: Improved logistic regression accuracy output.

## 7.9 Model Comparison

| Model                        | Predictors                                                                    | Test Accuracy |
|------------------------------|-------------------------------------------------------------------------------|---------------|
| Full Logistic Regression     | Gender, product line, unit price, quantity, rating, payment, month, day, hour | 57%           |
| Improved Logistic Regression | Gender, product line, quantity, payment, rating                               | 58%           |

A data.frame: 2 × 3

**Model Accuracy Accuracy\_Percentage**

**<chr>**

**<dbl>**

**<dbl>**

Full Logistic Regression

0.57

57

Improved Logistic Regression

0.58

58

Figure 29: Week 3 model comparison output.

The improved logistic regression model performed better on the test dataset even though the improved logistic regression model used a smaller set of predictors. This suggests that the additional time-related and unit-price variables included in the logistic regression model did not improve classification accuracy in this particular train-test split.

## 8. INTEGRATED RESULTS

Combining the three weeks gives a complete view of the supermarket sales data. The first week set up data quality for the supermarket sales data. The second week showed business patterns visually for the supermarket sales data. The third week tested selected relationships statistically. Evaluated customer-type classification for the supermarket sales data.

| Area                  | Integrated Finding                                                                                                                                               |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Quality          | No missing values or duplicate records were detected. Potential high-value outliers were reviewed and retained because they represented legitimate transactions. |
| Data Preparation      | Column names/types were standardized; sales normalization, categorical encoding, and time-based feature engineering were completed.                              |
| Sales Distribution    | Mean sales were 322.97 and median sales were 253.85, with most transactions concentrated below approximately 1,000.                                              |
| Product Performance   | Total sales varied across product lines, with the highest at 56,145 and the lowest at 49,194.                                                                    |
| Branch Performance    | Giza had the highest total sales and highest average customer rating in the Week 2 analysis.                                                                     |
| Time Pattern          | The highest sales intensity occurred on Tuesday at 19:00, while a relatively low period occurred on Monday at 14:00.                                             |
| Monthly Trend         | January had the highest total sales at 116,292, and February the lowest at 97,216.                                                                               |
| Customer Satisfaction | Food and beverages had the highest average rating at 7.11; Home and lifestyle had the lowest at 6.84.                                                            |
| Customer Type         | Member customers had the higher mean sales value and slightly greater variation in transaction values.                                                           |
| Statistical Testing   | Customer type was significantly associated with gender, but no significant association was detected with payment method or mean quantity purchased.              |
| Predictive Model      | The full logistic model achieved 57% accuracy; the reduced model achieved 58%.                                                                                   |
| Cross-Validation      | Mean ten-fold cross-validation accuracy was 56.75% for the full model.                                                                                           |

## 8.1 Main Analytical Story

Looking at the data, I realized that supermarket performance does not rely on one thing. Many different things change how a supermarket performs. Things like product category, branch, time, month, customer type, and customer satisfaction all show us sides of how people shop. I used analysis to find patterns, and then I used statistical analysis to make sure those patterns were actually real and not just accidents.

The predictive modeling results show that the details in a transaction can tell us a bit about customer type, but the results are only okay. The improved model has a 58% accuracy, which's only a tiny bit better than the 57% accuracy of the full model. This tells me that I need to find details and try different algorithms if I want the predictive modeling to work much better.

## 9. DISCUSSION AND BUSINESS IMPLICATIONS

The findings can be translated into several practical business considerations. The purpose of these implications is not to claim that the analysis proves a specific business action, but to show how the results could inform further decision-making.



## **9.1 Product and Inventory Management**

Product-line sales were not equal. Higher-performing product categories contributed more to overall revenue and may require closer inventory monitoring. Lower-performing categories can be investigated to understand whether the difference is related to customer demand, product placement, pricing, or other factors.

## **9.2 Branch Management**

The branch analysis showed the importance of combining sales, customer satisfaction, and transaction volume. Giza recorded the highest total sales and average customer rating in the Week 2 analysis, suggesting a useful benchmark for comparing branch performance.

## **9.3 Staffing and Service Planning**

The day-and-hour heatmap identified Tuesday at 19:00 as the highest sales-intensity period. Such information can support staff scheduling and service preparation during high-demand periods.

## **9.4 Seasonal and Monthly Planning**

Monthly sales fluctuated, with January recording the highest sales and February the lowest. Monitoring such changes can support monthly sales targets, inventory planning, and promotional decisions.

## **9.5 Customer Relationship Management**

Member customers showed a higher mean transaction sales value than Normal customers. Combined with the statistically significant association between customer type and gender, this suggests that customer segmentation could be explored further. However, the statistical relationship should not be interpreted as evidence that gender causes customer membership.

## **9.6 Predictive Model Use**

The classification model should currently be considered an analytical and exploratory tool rather than a production decision system. The 57% full-model accuracy and 58% reduced-model accuracy indicate that the available predictors provide only moderate predictive information about customer type. Further modeling would be necessary before operational use.

# **10. CHALLENGES FACED**

The project had hands-on challenges over the course of four weeks. When it came to preparing the data, there was a need to check what types of variables were being used, make sure the column names were all the same, tell the high-value transactions apart from the strange outliers, and get the categorical variables ready for the analysis that would come later.

When it came to making visuals, another challenge was picking the kinds of charts to show the message clearly. Different questions needed ways to show the data, like histograms to show how things are spread out, lollipop charts to compare categories, bubble charts to look at different branches with many factors, heatmaps to see patterns over time, line charts to track what happened month by month, and violin plots to look at the different types of customers.

The statistical part needed choosing the tests and understanding what the results really meant. The predictive modeling part needed being careful about which variable to focus on, making sure not to use future information, splitting the data into training and testing sets, understanding the confusion matrix using cross-validation, doing a diagnostic check, and comparing a full model with a simpler one.

These problems were solved by looking at the results, at every step making the process better as we went along, and thinking about what the results meant for the business question of just looking at the numbers.

## **11. LESSONS LEARNED**

- Data cleaning is the foundation of reliable analysis. A model or visualization is only as useful as the data supplied to it.
- Visualizations should be selected according to the analytical question and the type of variables being studied.
- Statistical significance should be interpreted using hypotheses, p-values, and the context of the test rather than by looking only at visible patterns.
- Classification models should be evaluated using multiple measures. Accuracy alone does not show how performance differs between classes.
- Cross-validation and diagnostics provide additional information about model stability and fit.
- Simpler models can sometimes perform as well as or better than more complex predictor sets.
- Business interpretation is an essential final step: analytical results become more useful when they are connected to decisions involving products, branches, customers, staffing, and planning.
- R provides a practical environment for carrying out a complete data-analysis workflow from preparation through communication.

## **12. RECOMMENDATIONS AND FUTURE SCOPE**

The current project provides a strong foundation, but several extensions could improve both analytical depth and predictive performance.

- Compare logistic regression with decision trees, random forests, and other suitable classification algorithms.
- Perform more extensive feature selection and feature engineering using transaction-level and customer-related variables.
- Test alternative probability thresholds to study the trade-off between sensitivity and specificity.
- Use a larger or more diverse dataset to improve generalizability.
- Perform repeated or stratified cross-validation to obtain a more robust estimate of predictive performance.
- Investigate customer segmentation to identify groups with different purchasing behavior.
- Extend the monthly analysis into time-series forecasting if a longer time span becomes available.
- Study the relationship between unit price, quantity, and sales in more depth.
- Develop an interactive dashboard for decision-makers using the final insights.
- Investigate why the reduced model slightly outperformed the full model and whether additional predictors improve performance consistently across validation samples.

## 13. CONCLUSION

This final report brings together everything I did over four weeks of analyzing the Supermarket Sales Dataset using R. I started by getting the data ready, then I moved on to making charts to show what I found. I ended by running statistical tests and building a predictive model.

During the data preparation stage, I checked the Supermarket Sales Dataset. Found 1,000 records and 17 variables. There were no missing values or duplicate records. I looked for outliers. Kept them if they were real transactions. I also used normalization, encoding, and feature engineering to get the Supermarket Sales Dataset ready for the next steps.

When I looked at the charts, I saw some big patterns. The mean transaction sales were 322.97. The median sales were 253.85. I noticed that sales for each product line were different, and each branch had sales and ratings. Sales were highest on Tuesday at 19:00. January had the highest monthly sales, and customer satisfaction changed depending on the product line. I also saw that Member customers had a higher mean sales value than Normal customers.

The statistical tests gave me proof. The Quantity, unit price, and rating did not pass the Shapiro-Wilk normality test. A t-test showed there was no difference in mean quantity between the different customer types. I found that customer type was linked to gender. Customer type had no real link to the payment method used.

For this stage, I used logistic regression to try and guess the customer type. The full model had a 57% test accuracy, 75.63% sensitivity, and 29.63% specificity. When I used ten-fold cross-validation, the mean accuracy was 56.75%. I also tried a reduced regression model, which had a 58% test accuracy and worked a little better than the full model.

Looking back, this project shows why a full data approach is so useful. Cleaning the data makes sure it is good; visualization helps show patterns; statistical testing proves what is happening, and predictive modeling helps with making decisions. My analysis shows that the current model still has limits. In the future, people should try algorithms, better features, and more testing.

This four-week internship gave me real-world experience. I learned more than R programming and statistical analysis. I learned how to understand results, how to explain what I found, and how to connect my findings to business decisions.

## 14. REFERENCES

1. Supermarket Sales Dataset. Transaction-level supermarket sales data used throughout the four-week internship project.
2. R programming language and standard R statistical functions used for data preparation, statistical analysis, and modeling.
3. ggplot2, dplyr, and scales packages used for data visualization and data manipulation.
4. Week 1 Report: Data Cleaning and Preliminary Analysis using R, submitted 29 July 2026.
5. Week 2 Report: Data Visualization and Insight Communication using R, submitted 6 August 2026.
6. Week 3 Report: Statistical Analysis and Predictive Modeling using R, submitted 16 August 2026.

## 15. APPENDIX

The appendix contains selected supporting outputs from the three previous weeks. The main body of the report includes the most important figures and results so that the final report remains readable while still demonstrating the technical work completed.

### 15.1 Selected Week 1 Supporting Outputs

```
[27]: head(sales[, c(
      "date",
      "time",
      "month",
      "day",
      "hour",
      "time_period"
    )])
```

A data.frame: 6 × 6

|   | date       | time        | month | day      | hour  | time_period |
|---|------------|-------------|-------|----------|-------|-------------|
|   | <date>     | <chr>       | <ord> | <chr>    | <int> | <chr>       |
| 1 | 2019-01-05 | 1:08:00 PM  | Jan   | Saturday | 13    | Afternoon   |
| 2 | 2019-03-08 | 10:29:00 AM | Mar   | Friday   | 10    | Morning     |
| 3 | 2019-03-03 | 1:23:00 PM  | Mar   | Sunday   | 13    | Afternoon   |
| 4 | 2019-01-27 | 8:33:00 PM  | Jan   | Sunday   | 20    | Evening     |
| 5 | 2019-02-08 | 10:37:00 AM | Feb   | Friday   | 10    | Morning     |
| 6 | 2019-03-25 | 6:30:00 PM  | Mar   | Monday   | 18    | Evening     |

```
[28]: sum(is.na(sales$hour))
0
```

Appendix Figure A1: Week 1 dataset inspection/data preparation output.

```
[20]: library(lubridate)

[21]: # Create Month
sales$month <- month(sales$date, label = TRUE)

# Create Day of Week
sales$day <- weekdays(sales$date)

[22]: # Convert time column to character
sales$time <- as.character(sales$time)

# Display sample values
head(sales$time)

'1:08:00 PM' · '10:29:00 AM' · '1:23:00 PM' · '8:33:00 PM' · '10:37:00 AM' · '6:30:00 PM'

[23]: # Extract hour from the time column

sales$hour <- suppressWarnings(
  as.numeric(substr(trimws(sales$time), 1, 2))
)

# Check the result
head(sales$hour)

<NA> · 10 · <NA> · <NA> · 10 · <NA>

[24]: sum(is.na(sales$hour))

720
```

Appendix Figure A2: Week 1 categorical encoding output.

## 15.2 Selected Week 3 Supporting Outputs

Call:

```
glm(formula = customer_type ~ gender + product_line + unit_price +  
  quantity + rating + payment + month + day + hour, family = binomial,  
  data = train_data)
```

Coefficients:

|                                 | Estimate  | Std. Error | z value | Pr(> z )     |
|---------------------------------|-----------|------------|---------|--------------|
| (Intercept)                     | 0.194910  | 0.588276   | 0.331   | 0.740400     |
| genderMale                      | 0.558349  | 0.149418   | 3.737   | 0.000186 *** |
| product_lineFashion accessories | 0.043168  | 0.249062   | 0.173   | 0.862399     |
| product_lineFood and beverages  | -0.185596 | 0.247755   | -0.749  | 0.453788     |
| product_lineHealth and beauty   | -0.034429 | 0.262456   | -0.131  | 0.895632     |
| product_lineHome and lifestyle  | -0.115349 | 0.261135   | -0.442  | 0.658691     |
| product_lineSports and travel   | -0.113935 | 0.256704   | -0.444  | 0.657160     |
| unit_price                      | -0.004880 | 0.002787   | -1.751  | 0.079995 .   |
| quantity                        | -0.030832 | 0.025648   | -1.202  | 0.229321     |
| rating                          | 0.022432  | 0.043357   | 0.517   | 0.604886     |
| paymentCredit card              | -0.262909 | 0.183115   | -1.436  | 0.151071     |
| paymentEwallet                  | 0.116437  | 0.175484   | 0.664   | 0.506996     |
| month.L                         | -0.005246 | 0.125449   | -0.042  | 0.966642     |
| month.Q                         | 0.176089  | 0.129891   | 1.356   | 0.175203     |
| dayMonday                       | 0.235427  | 0.279232   | 0.843   | 0.399159     |
| daySaturday                     | -0.009235 | 0.268073   | -0.034  | 0.972519     |
| daySunday                       | -0.133415 | 0.283963   | -0.470  | 0.638474     |
| dayThursday                     | 0.089505  | 0.277847   | 0.322   | 0.747350     |
| dayTuesday                      | -0.462194 | 0.276827   | -1.670  | 0.094996 .   |
| dayWednesday                    | 0.005762  | 0.277451   | 0.021   | 0.983432     |
| hour                            | -0.016773 | 0.023184   | -0.723  | 0.469397     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1098.4 on 799 degrees of freedom  
Residual deviance: 1061.2 on 779 degrees of freedom  
AIC: 1103.2

Number of Fisher Scoring iterations: 4

*Appendix Figure A3: Week 3 statistical/model output.*

```

# Part 14 – Train/Test Split

set.seed(123)

# Create random training indices
train_index <- sample(
  1:nrow(model_data),
  size = 0.80 * nrow(model_data)
)

# Create training data
train_data <- model_data[train_index, ]

# Create testing data
test_data <- model_data[-train_index, ]

# Check sizes
nrow(train_data)
nrow(test_data)

```

800

200

*Appendix Figure A4: Train-test split output.*

```

[79]: mean_cv_accuracy <- mean(cv_accuracy)
      mean_cv_accuracy
      0.5675

[80]: mean_cv_accuracy_percentage <- mean_cv_accuracy * 100
      mean_cv_accuracy_percentage
      56.75

R11

```

*Appendix Figure A5: Cross-validation summary.*

All figures and results presented in this final report were taken from the work completed during Weeks 1–3. No new analytical coding or dataset was introduced for Week 4; the final week focused on integration, interpretation, reporting, and presentation.